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Project Name : **Exploratory Data Analysis and Customer Churn Modeling for E-Commerce Customer Insights Using Python**

Project Domain : **E-commerce**

Submission Date : 1/04/2026

Mentor Name : Kumaran M

Raw Dataset Link : https://drive.google.com/file/d/1Qr19u3lekWE0ocEIVud266u6Rp67XS1-/view?usp=sharing

Cleaned Dataset link : https://drive.google.com/file/d/1v3Grq3x4bOCPKCUJD1Bh6YO-59e4uvqw/view?usp=drive_link

Project Title

Exploratory Data Analysis and Customer Churn Modeling for E-Commerce Customer Insights Using Python

Domain: E-commerce

Objective:

1. To explore and understand the e-commerce dataset using Exploratory Data Analysis (EDA).

2. To analyze customer behavior and purchase patterns such as product categories, order value, and country distribution.

3. To identify factors that influence customer churn based on customer activity and subscription status.

4. To provide actionable insights and recommendations from the analysis

Outcome:

To identify the key factors influencing customer churn and predict whether a customer is likely to remain active or leave the e-commerce platform**.**

Dataset Information:

- Source: Source: Kaggle – E-Commerce Customer Insights and Churn Dataset

- Year / Timeline: Data collected during 2021 to 2025

- Dataset Description: This dataset contains 2,000 customer-level and order-level records from a simulated e-commerce platform. It includes

information related to customer demographics, purchasing behaviour, order history, and churn status.

- Order Id - Unique identifier for each order placed on the platform.
- Customer Id - Unique identifier assigned to each customer.
- Customer Age - Age of the customer who placed the order.
- product_id - Unique identifier for the purchased product.
- country - Country where the customer is located.
- signup date - Date when the customer registered on the platform.
- last_purchase_date - The most recent purchase date of the customer before the current order.
- cancellations count - Total number of orders cancelled by the customer.
- subscription status - Current subscription status of the customer (active, paused, cancelled).
- order date - Date when the order was placed.
- unit price - Price of a single unit of the product.
- quantity - Number of units purchased in the order.
- purchase count - Total number of purchases made by the customer.
- preferred category - Product category most frequently purchased by the customer.

- product name - Name of the product purchased.
- category - Product category (Electronics, Clothing, Sports, Beauty, etc.)
- gender - Gender of the customer.

Type of Analysis

1. Exploratory Data Analysis (EDA) is used to understand the dataset, identify patterns, and analyse customer behaviour.
2. Predictive Analysis is used to build a model that predicts whether a customer is likely to churn or remain active.

Stages for DA Project

- Stage 1 – Problem Definition and Dataset Selection
- Stage 2 – Data Cleaning and Pre-processing
- Stage 3 – EDA and Visualizations
- Stage 4 – Documentation, Insights and Presentation

Stage 1 – Problem Definition and Dataset Selection

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/E Commerce Customer Insights and Churn Dataset.csv")

df.head()
```

sample output:

	order_id	customer_id	age	product_id	country	signup_date	last_purchase_date	cancellations_count	subscription_status	order_date	unit_price	quantity	purchase_frequency	preferred_category	product_name	category	gender
0	ORD5000	CUST1000	39	PROD200	Canada	1/7/2021	2/21/2023	0	active	8/20/2024	78.21	5	37	Sports	Football	Sports	Female
1	ORD5001	CUST1001	61	PROD201	USA	10/19/2020	12/8/2021	0	active	7/17/2025	64.02	8	35	Sports	Refrigerator	Home	Female
2	ORD5002	CUST1002	26	PROD202	Pakistan	6/10/2023	9/4/2023	3	cancelled	3/12/2025	604.14	2	44	Electronics	Hoodie	Clothing	Female
3	ORD5003	CUST1003	54	PROD203	India	7/30/2023	2/20/2024	4	paused	9/19/2024	84.66	4	1	Sports	Conditioner	Beauty	Male
4	ORD5004	CUST1004	50	PROD204	India	12/9/2020	9/14/2024	0	active	8/8/2024	62.66	7	35	Sports	Smartwatch	Electronics	Male

```
print(df.head)
```

sample output :

```
<bound method NDFrame.head of      order_id customer_id  age product_id   country signup_date  \
0     ORD5000    CUST1000   39    PROD200    Canada    1/7/2021
1     ORD5001    CUST1001   61    PROD201       USA  10/19/2020
2     ORD5002    CUST1002   26    PROD202  Pakistan   6/10/2023
3     ORD5003    CUST1003   54    PROD203     India   7/30/2023
4     ORD5004    CUST1004   50    PROD204     India   12/9/2020
...       ...         ...  ...        ...       ...         ...
1995  ORD6995    CUST2995   27   PROD2195        UK   7/11/2021
1996  ORD6996    CUST2996   59   PROD2196    Canada  11/26/2022
1997  ORD6997    CUST2997   41   PROD2197     India   4/20/2022
1998  ORD6998    CUST2998   20   PROD2198       USA   5/16/2021
1999  ORD6999    CUST2999   54   PROD2199     India   1/31/2023

     last_purchase_date  cancellations_count subscription_status order_date  \
0            2/21/2023                    0              active  8/20/2024
1            12/8/2021                    0              active  7/17/2025
2             9/4/2023                    3           cancelled  3/12/2025
3            2/20/2024                    4              paused  9/19/2024
4            9/14/2024                    0              active   8/8/2024
...                ...                  ...                 ...        ...
1995         7/22/2023                    0              active  7/23/2022
1996        10/29/2023                    4              paused   3/5/2024
1997         8/15/2022                    1              active  4/29/2023
1998        12/15/2024                    0              active  2/17/2022
1999         7/25/2023                    5           cancelled  5/31/2024

      unit_price  quantity  purchase_frequency preferred_category  \
0          78.21         5                  37            Sports
1          64.02         8                  35            Sports
2         604.14         2                  44       Electronics
3          84.66         4                   1            Sports
4          62.66         7                  35            Sports
...          ...       ...                 ...               ...
1995       30.34         7                  25            Beauty
1996      381.38         3                  12       Electronics
1997      305.05         6                  38            Beauty
1998      229.14         1                  49            Beauty
1999        4.41         7                  24            Beauty

      product_name     category  gender
0         Football       Sports  Female
1     Refrigerator         Home  Female
2           Hoodie     Clothing  Female
3      Conditioner       Beauty    Male
4       Smartwatch  Electronics    Male
...            ...          ...     ...
1995      Sneakers     Clothing  Female
1996         Chair         Home    Male
1997      Lipstick       Beauty    Male
1998     Face Wash       Beauty    Male
1999        Hoodie     Clothing    Male

[2000 rows x 17 columns]>
```

```
print(df.describe)
```

sample output :

```
<bound method DataFrame.info of      order_id customer_id  age product_id   country signup_date  \
0     ORD5000    CUST1000   39    PROD200    Canada    1/7/2021
1     ORD5001    CUST1001   61    PROD201       USA  10/19/2020
2     ORD5002    CUST1002   26    PROD202  Pakistan   6/10/2023
3     ORD5003    CUST1003   54    PROD203     India   7/30/2023
4     ORD5004    CUST1004   50    PROD204     India   12/9/2020
...       ...         ...  ...        ...       ...         ...
1995  ORD6995    CUST2995   27   PROD2195        UK   7/11/2021
1996  ORD6996    CUST2996   59   PROD2196    Canada  11/26/2022
1997  ORD6997    CUST2997   41   PROD2197     India   4/20/2022
1998  ORD6998    CUST2998   20   PROD2198       USA   5/16/2021
1999  ORD6999    CUST2999   54   PROD2199     India   1/31/2023

     last_purchase_date  cancellations_count subscription_status order_date  \
0             2/21/2023                    0              active  8/20/2024
1             12/8/2021                    0              active  7/17/2025
2              9/4/2023                    3           cancelled  3/12/2025
3             2/20/2024                    4              paused  9/19/2024
4             9/14/2024                    0              active   8/8/2024
...                 ...                  ...                 ...        ...
1995          7/22/2023                    0              active  7/23/2022
1996         10/29/2023                    4              paused   3/5/2024
1997          8/15/2022                    1              active  4/29/2023
1998         12/15/2024                    0              active  2/17/2022
1999          7/25/2023                    5           cancelled  5/31/2024

      unit_price  quantity  purchase_frequency preferred_category  \
0          78.21         5                  37             Sports
1          64.02         8                  35             Sports
2         604.14         2                  44        Electronics
3          84.66         4                   1             Sports
4          62.66         7                  35             Sports
...          ...       ...                 ...                ...
1995       30.34         7                  25             Beauty
1996      381.38         3                  12        Electronics
1997      305.05         6                  38             Beauty
1998      229.14         1                  49             Beauty
1999        4.41         7                  24             Beauty

      product_name     category  gender
0         Football       Sports  Female
1     Refrigerator         Home  Female
2           Hoodie     Clothing  Female
3      Conditioner       Beauty    Male
4       Smartwatch  Electronics    Male
...            ...          ...     ...
1995      Sneakers     Clothing  Female
1996         Chair         Home    Male
1997      Lipstick       Beauty    Male
1998     Face Wash       Beauty    Male
1999        Hoodie     Clothing    Male

[2000 rows x 17 columns]>
```

print(df.shape)
(2000,17)

sample output :

```
(2000, 17)
(2000, 17)
```

print(df.dtypes)

sample output :

```
order_id                object
customer_id             object
age                      int64
product_id              object
country                 object
signup_date             object
last_purchase_date      object
cancellations_count      int64
subscription_status     object
order_date              object
unit_price             float64
quantity                 int64
purchase_frequency       int64
preferred_category      object
product_name            object
category                object
gender                  object
dtype: object
```

print(df.isnull().sum())

sample output :

```
order_id                object
customer_id             object
age                      int64
product_id              object
country                 object
signup_date             object
last_purchase_date      object
cancellations_count      int64
subscription_status     object
order_date              object
unit_price             float64
quantity                 int64
purchase_frequency       int64
preferred_category      object
product_name            object
category                object
gender                  object
dtype: object
```

```
print(df.duplicated().sum)
```

sample output :

```
<bound method Series.sum of 0       False
1       False
2       False
3       False
4       False
        ...
1995    False
1996    False
1997    False
1998    False
1999    False
Length: 2000, dtype: bool>
```

Stage 2 – Data Cleaning and Pre-processing

Clean column names

```
df.columns = df.columns.str.strip()
```

Check missing values

```
df.isnull().sum()
```

sample output :

	0
order_id	0
customer_id	0
age	0
product_id	0
country	0
signup_date	0
last_purchase_date	0
cancellations_count	0
subscription_status	0
order_date	0
unit_price	0
quantity	0
purchase_frequency	0
preferred_category	0
product_name	0
category	0
gender	0

dtype: int64

Handle duplicates

```
df.duplicated().sum()
```

sample output :

```
np.int64(0)
```

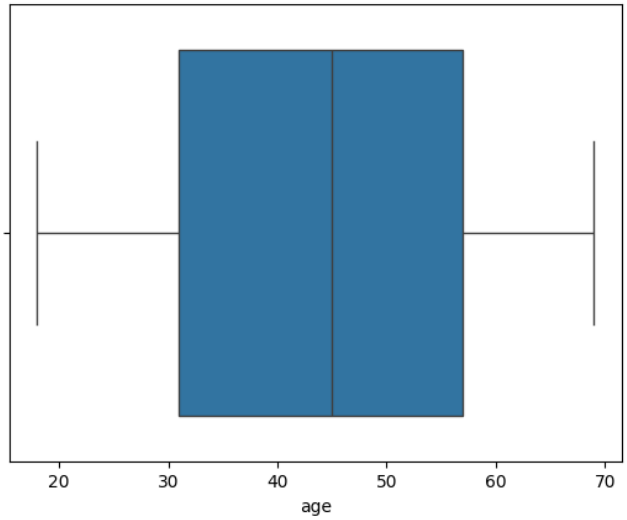
Convert date columns

```
df["signup_date"] = pd.to_datetime(df["signup_date"])
df["last_purchase_date"] = pd.to_datetime(df["last_purchase_date"])
df["order_date"] = pd.to_datetime(df["order_date"])
```

Treat Outliers

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.boxplot(x=df["age"])
plt.show()
sns.boxplot(x=df["unit_price"])
plt.show()
sns.boxplot(x=df["quantity"])
plt.show()
```

sample output :



Check skewness

```
df[["age","unit_price","quantity"]].skew()
```

sample output :

	0
age	-0.102464
unit_price	2.392840
quantity	0.033510

dtype: float64

```
df["total_order_value"] = df["unit_price"] * df["quantity"]

display(df)
```

sample output :

	order_id	customer_id	age	product_id	country	signup_date	last_purchase_date	cancellations_count	subscription_status	order_date	unit_price	quantity	purchase_frequency	preferred_category	product_name	category	gender
0	ORD5000	CUST1000	39	PROD200	Canada	2021-01-07	2023-02-21	0	active	2024-08-20	78.21	5	37	Sports	Football	Sports	Female
1	ORD5001	CUST1001	61	PROD201	USA	2020-10-19	2021-12-08	0	active	2025-07-17	64.02	8	35	Sports	Refrigerator	Home	Female
2	ORD5002	CUST1002	26	PROD202	Pakistan	2023-06-10	2023-09-04	3	cancelled	2025-03-12	604.14	2	44	Electronics	Hoodie	Clothing	Female
3	ORD5003	CUST1003	54	PROD203	India	2023-07-30	2024-02-20	4	paused	2024-09-19	84.66	4	1	Sports	Conditioner	Beauty	Male
4	ORD5004	CUST1004	50	PROD204	India	2020-12-09	2024-09-14	0	active	2024-08-08	62.66	7	35	Sports	Smartwatch	Electronics	Male
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1995	ORD6995	CUST2995	27	PROD2195	UK	2021-07-11	2023-07-22	0	active	2022-07-23	30.34	7	25	Beauty	Sneakers	Clothing	Female
1996	ORD6996	CUST2996	59	PROD2196	Canada	2022-11-26	2023-10-29	4	paused	2024-03-05	381.38	3	12	Electronics	Chair	Home	Male
1997	ORD6997	CUST2997	41	PROD2197	India	2022-04-20	2022-08-15	1	active	2023-04-29	305.05	6	38	Beauty	Lipstick	Beauty	Male
1998	ORD6998	CUST2998	20	PROD2198	USA	2021-05-16	2024-12-15	0	active	2022-02-17	229.14	1	49	Beauty	Face Wash	Beauty	Male
1999	ORD6999	CUST2999	54	PROD2199	India	2023-01-31	2023-07-25	5	cancelled	2024-05-31	4.41	7	24	Beauty	Hoodie	Clothing	Male

2000 rows × 17 columns

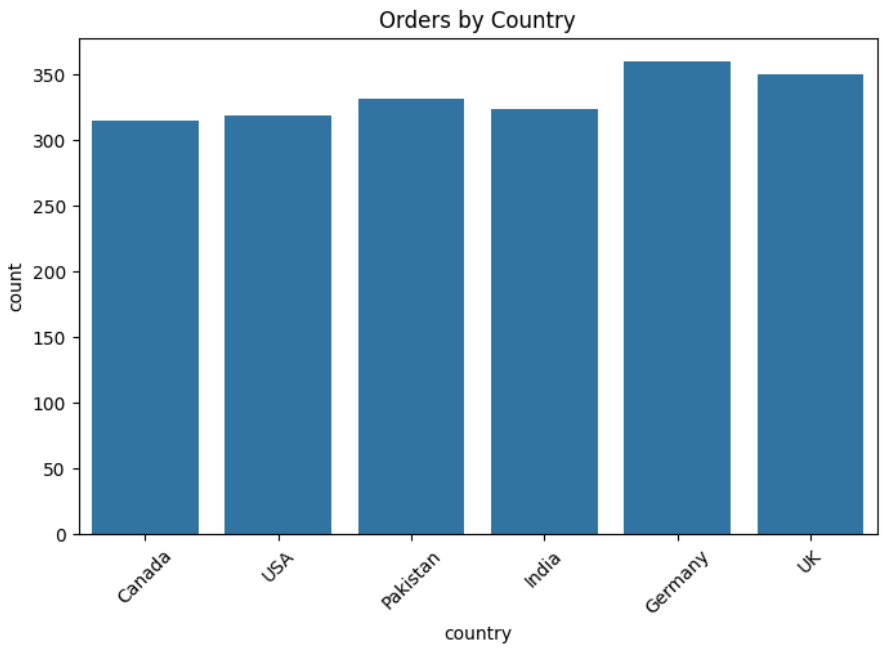
Cleaned Dataset

```
df.to_csv("cleaned_dataset_E_commerce_.csv", index = False)
```

Stage 3 – EDA and Visualizations

1 Countplot – Orders by Country

```
plt.figure(figsize=(8,5))
sns.countplot(x="country", data=df)
plt.title("Orders by Country")
plt.xticks(rotation=45)
plt.show()
```



Interpretation of 1.Orders by Country (Countplot)

- The above countplot visualizes the distribution of customer orders across different countries.
- The chart compares how many orders were placed by customers from each country, showing which regions contribute more to the platform's overall activity.

Features

- Several key visual features are used in this chart:
- X-axis (Country): Represents the different countries where customers are located.
- Y-axis (Count): Represents the number of orders placed by customers.

Bars: Each bar shows the total number of orders from a particular country.

Insights

- Some countries have significantly higher order counts than others.
- This indicates stronger customer engagement in those regions.
- Businesses can use this insight to focus marketing strategies on high-performing countries

2.Countplot - Orders by Category

```
plt.figure(figsize=(8,5))
sns.countplot(x="category", data=df)
plt.title("Orders by Product Category")
plt.xticks(rotation=45)
plt.show()
```

Interpretation of Chart 2 – Orders by Product Category (Countplot)

- The above countplot visualizes the number of orders placed across different product categories.
- It helps compare which categories customers purchase most frequently.

Features

- X-axis (Category): Represents different product categories available on the platform.
- Y-axis (Count): Represents the total number of orders placed in each category.

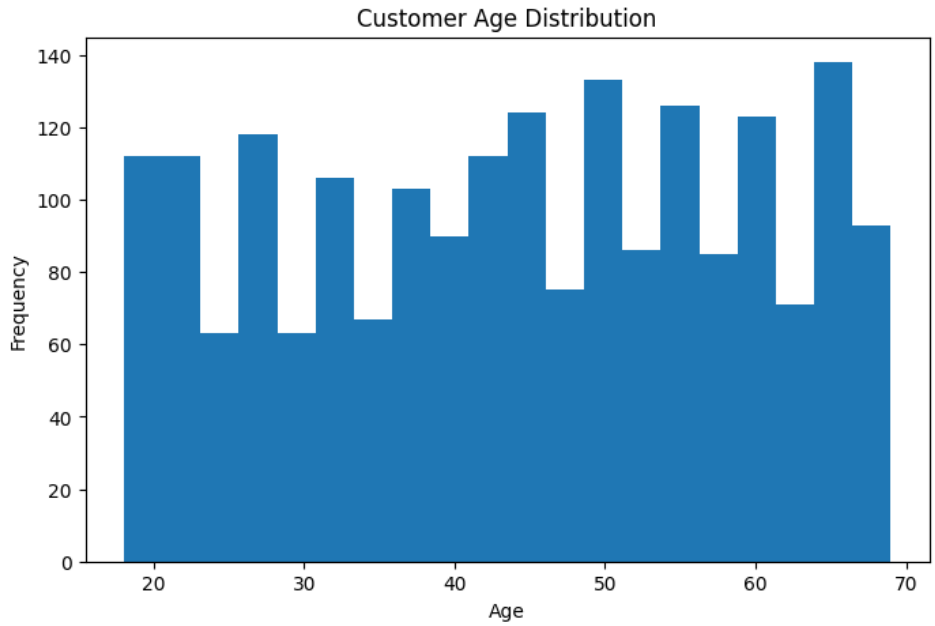
Bars: Show the popularity of each product category.

Insights

- Some product categories receive more orders than others.
- High-demand categories can be prioritized for inventory and promotions.

3 Histogram (Age Distribution)

```
plt.figure(figsize=(8,5))
plt.hist(df["age"], bins=20)
plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



Interpretation of Chart 3 – Customer Age Distribution (Histogram)

- The above histogram visualizes the distribution of customers based on their age.
- It helps understand which age groups are most active on the e-commerce platform.

Features

- X-axis (Age): Represents customer age groups.
- Y-axis (Frequency): Represents the number of customers within each age range.

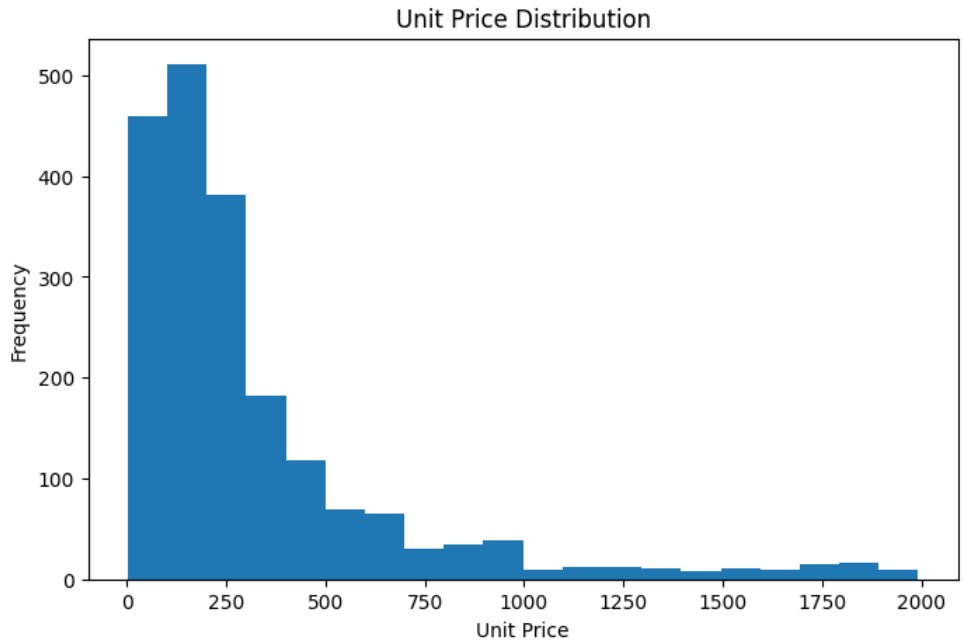
Bars: Show how many customers fall within each age interval.

Insights

- Most customers fall within a specific age range.
- Businesses can target marketing strategies toward these dominant age groups.

4 Histogram (Unit Price Distribution)

```
plt.figure(figsize=(8,5))
plt.hist(df["unit_price"], bins=20)
plt.title("Unit Price Distribution")
plt.xlabel("Unit Price")
plt.ylabel("Frequency")
plt.show()
```



Interpretation of yourChart 4 – Unit Price Distribution (Histogram)

- The above histogram visualizes the distribution of product prices across the platform.
- It helps understand the range and frequency of product pricing.

Features

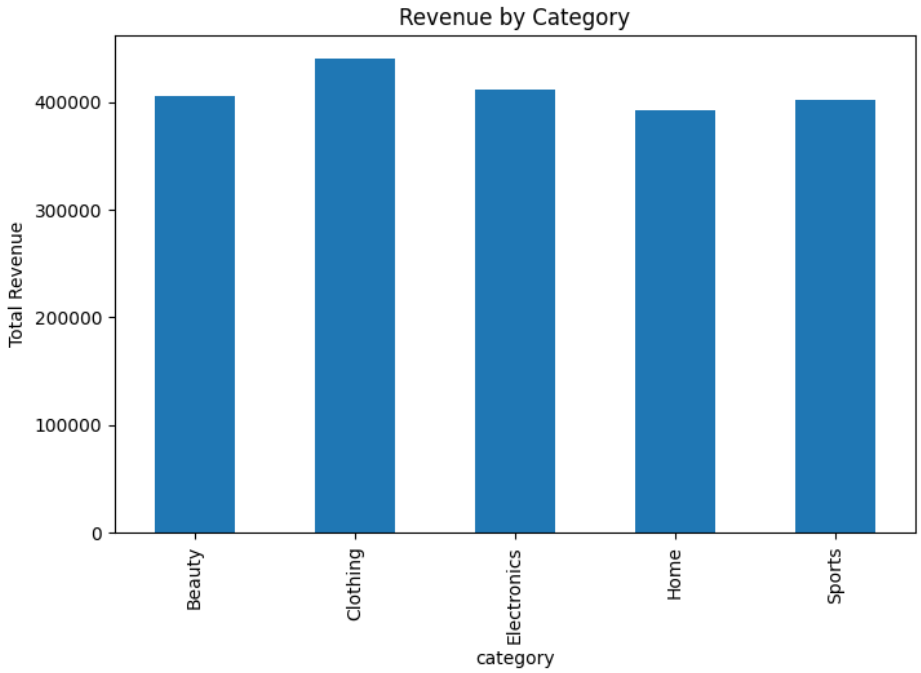
- X-axis (Unit Price): Represents product price ranges.
- Y-axis (Frequency): Represents the number of products within each price range.

Insights

- Most products fall within a certain price range.
- Extremely high-priced products appear less frequently.

5 Bar Chart (Revenue by Category)

```
category_revenue = df.groupby("category")["total_order_value"].sum()
category_revenue.plot(kind="bar", figsize=(8,5))
plt.title("Revenue by Category")
plt.ylabel("Total Revenue")
plt.show()
```



Interpretation of Chart 5 – Revenue by Category (Bar Chart)

- The above bar chart shows the total revenue generated by each product category.
- It compares which categories contribute the most revenue to the platform.

Features

- X-axis (Category): Represents different product categories.
- Y-axis (Total Revenue): Represents the sum of total order value for each category.

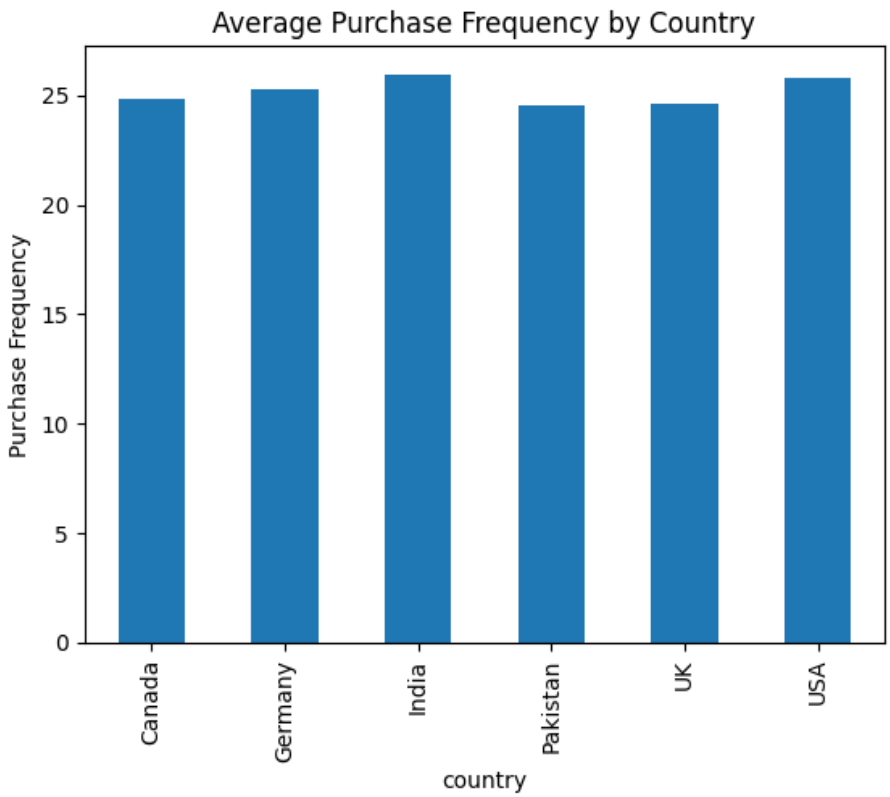
Bars: Show revenue contribution from each category.

Insights

- Some categories generate higher revenue than others.
- These categories play a major role in the company's overall sales performance.

6 Bar Chart (Purchase Frequency by Country)

```
country_purchase = df.groupby("country")["purchase_frequency"].mean()
country_purchase.plot(kind="bar")
plt.title("Average Purchase Frequency by Country")
plt.ylabel("Purchase Frequency")
plt.show()
```



Interpretation of Chart 6 – Purchase Frequency by Country (Bar Chart)

- The above bar chart visualizes the average purchase frequency of customers from different countries.

Features

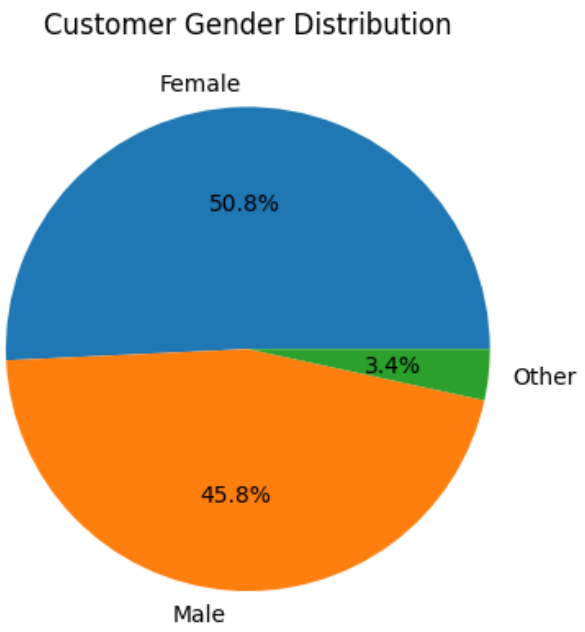
- X-axis (Country): Represents customer locations.
- Y-axis (Average Purchase Frequency): Represents how frequently customers make purchases.

Insights

- Customers from certain countries purchase more frequently.
- This indicates higher customer engagement in those regions.

7 — Pie Chart (Gender Distribution)

```
df["gender"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Customer Gender Distribution")
plt.ylabel("")
plt.show()
```



Interpretation of Chart 7 – Gender Distribution (Pie Chart)

- The above pie chart shows the proportion of male and female customers on the platform.

Features

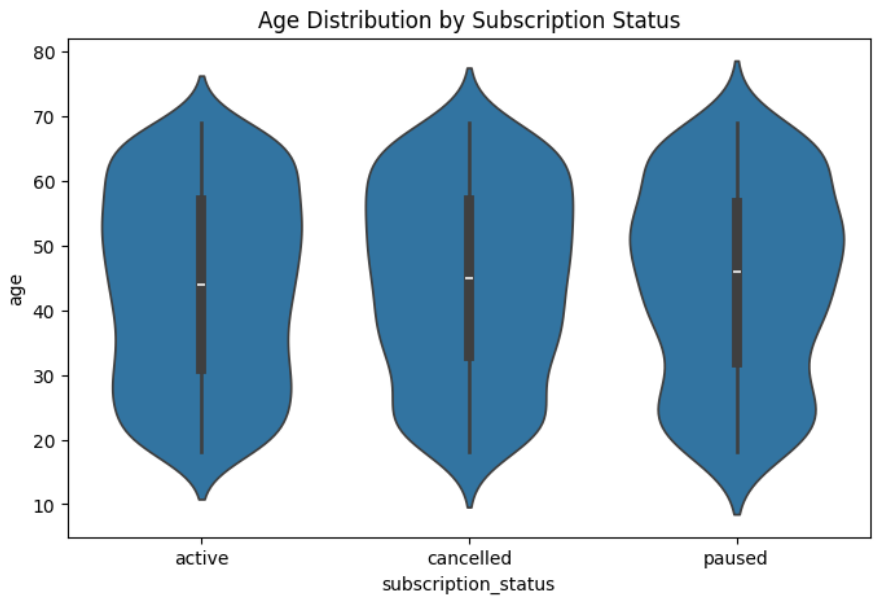
- Pie segments: Represent different gender groups.
- Percentage labels: Show the proportion of each gender.

Insights

- The chart reveals the gender distribution of the customer base.
- Businesses can analyze whether marketing strategies appeal equally to both genders.

8 — Violin Plot (Age vs Subscription Status)

```
plt.figure(figsize=(8,5))
sns.violinplot(x="subscription_status", y="age", data=df)
plt.title("Age Distribution by Subscription Status")
plt.show()
```



Interpretation of Chart 8 – Age vs Subscription Status (Violin Plot)

- The above violin plot visualizes the distribution of customer ages across different subscription statuses.

Features

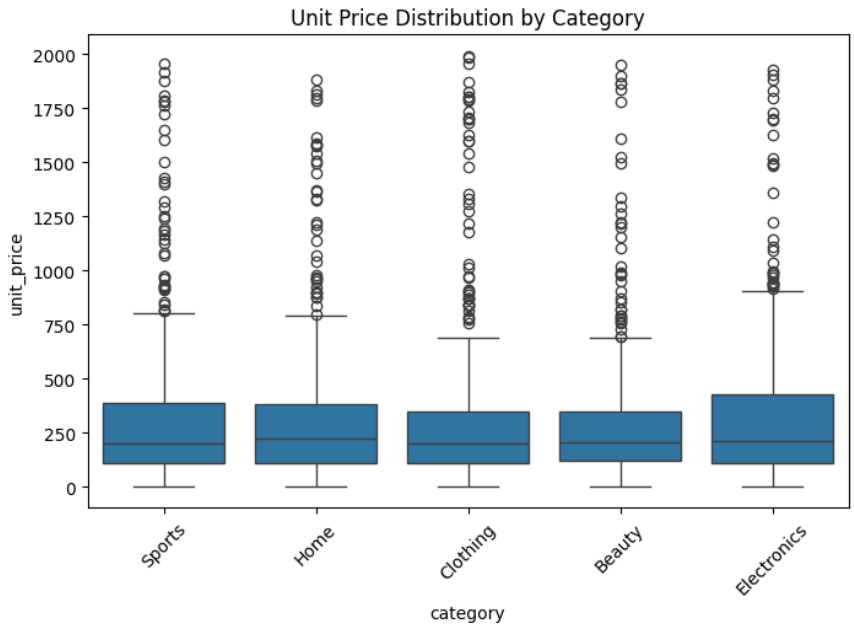
- X-axis (Subscription Status): Represents customer subscription categories such as active, paused, or cancelled.
- Y-axis (Age): Represents customer age distribution.
- Violin shapes: Show the density and spread of ages within each group.

Insights

- Different subscription statuses may be associated with different age groups.
- This helps identify patterns related to customer behavior.

9 — Box Plot (Unit Price by Category)

```
plt.figure(figsize=(8,5))
sns.boxplot(x="category", y="unit_price", data=df)
plt.title("Unit Price Distribution by Category")
plt.xticks(rotation=45)
plt.show()
```



Interpretation of Chart 9 – Unit Price by Category (Box Plot)

- The above box plot visualizes the distribution of product prices within each category.

Features

- X-axis (Category): Represents product categories.
- Y-axis (Unit Price): Represents product price values.
- Box and whiskers: Show median, quartiles, and outliers.

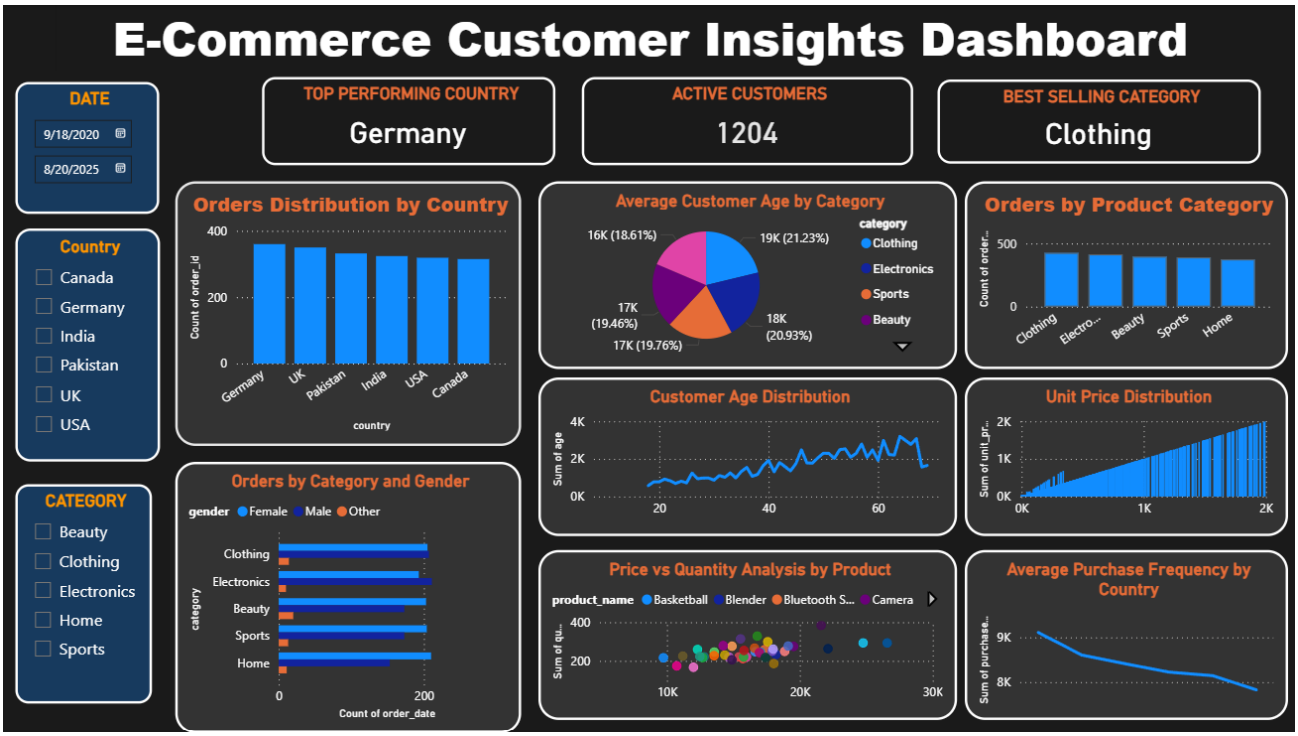
Insights

- Some categories contain higher-priced products than others.
- The presence of outliers indicates products priced significantly higher than the average.

Stage 4 – Documentation, Insights and Presentation

SUMMARY

- This dashboard analyzes customer behavior, product performance, and sales trends in an e-commerce platform.
- The data shows how different countries, product categories, and customer demographics contribute to overall business performance.
- We can observe patterns in customer purchasing behavior, pricing trends, and category-level demand, which helps understand where the business is performing well and where improvements are needed.



KEY INSIGHTS

1. Germany and the UK consistently appear as top-performing countries, indicating strong market demand in these regions.

2. Clothing and Electronics are the best-selling categories, contributing significantly to total orders and revenue.

3. Customer age shows a steady distribution, with most active buyers falling in the mid-age group (around 30–60), indicating a mature customer base.

4. Higher unit prices are associated with lower purchase frequency, suggesting price sensitivity among customers.

5. Purchase frequency varies by country, showing that some regions have more engaged and repeat customers than others.

BUSINESS RECOMMENDATIONS

1. Focus marketing and promotional campaigns in high-performing countries like Germany and the UK to maximize revenue.

2. Increase inventory and product variety in top categories such as Clothing and Electronics to meet demand.

3. Introduce targeted offers or discounts for high-priced products to improve purchase frequency.

4. Develop loyalty programs in countries with lower purchase frequency to increase customer engagement.

5. Use customer age insights to design age-specific marketing strategies for better conversion.

FINAL STORY

- This dashboard tells a complete story of how customers interact with the e-commerce platform.

- It shows that business performance is strongly driven by specific countries and product categories. While some regions contribute higher sales, others show opportunities for growth.

- Customer behavior analysis reveals that pricing plays a key role in purchasing decisions, and mid-aged customers form the majority of buyers.

- By combining these insights, the business can make data-driven decisions to improve sales, optimize pricing strategies, and target the right customers effectively.

- Overall, the dashboard provides a clear view of performance, customer trends, and actionable opportunities for business growth.

FUTURE ENHANCEMENTS

1. Expand Dataset Scope

Incorporate larger and more recent datasets, including real-time transactional data, to improve accuracy and uncover deeper business insights.

2. Automation & Pipeline Development

Automate data cleaning, transformation, and reporting processes using Python and scheduled workflows to reduce manual effort and improve efficiency.

3. Advanced Dashboard Interactivity

Enhance the dashboard by adding drill-down features, dynamic filters, and user-driven controls in Power BI to enable deeper exploration of insights.

4. Predictive Analytics Integration

Extend the analysis by implementing machine learning models to forecast sales, customer churn, and demand trends for proactive decision-making.

5. Customer Segmentation & Personalization

Apply clustering techniques to segment customers based on behavior and purchasing patterns, enabling targeted marketing strategies.

CONCLUSION

- This project analyzed customer behavior, product performance, and sales trends within an e-commerce platform to uncover key business insights. The analysis identified that specific countries and product categories significantly drive overall performance, while customer demographics and pricing directly influence purchasing behavior.

- The dashboard highlights important patterns such as high-performing regions, best-selling categories, and the relationship between price and demand. These insights provide a clear understanding of where the business is succeeding and where improvements can be made.

- By leveraging these findings, businesses can make informed decisions related to marketing strategies, pricing optimization, and customer engagement to improve overall performance.

- Ultimately, this project demonstrates how data-driven analysis can transform raw data into actionable insights that support smarter and more effective business decisions.